

User Guide

Beyond Mean-Variance Portfolio Optimiser

With Derivatives & Structured Products — A Mental Accounts Framework

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 sami-jeddou-behavioral-portfolio-optimizer.streamlit.app

Overview



Interactive Frontiers

MV vs Behavioural vs Derivative frontier — visualised side by side on a single chart



Three Portfolios

No derivative / With derivative / Same risk — three perspectives for each optimisation run



AI-Powered

On-demand explanations for every parameter, constraint, derivative type, and result

Three Portfolio Perspectives

1

Portfolio (1) — No derivative:

Equivalent to MV optimum — confirms MVT/MAT equivalence. Most classical optimisers stop here.

2

Portfolio (2) — With derivative:

Same implied λ and downside constraint (H, α or H, α, L) — shows return uplift from derivatives.

3

Portfolio (3) — Same risk as (1):

Interpolated at $P(1)$ std dev — direct comparison of return gain at equivalent risk (indicative).

1

Step 1 — Portfolio Data

1

PORTFOLIO DATA

☐ Default (3-asset sample case)

☒ Live market data (Yahoo Finance)

☐ Enter manually

☐ Upload CSV

Ticker symbols (comma-separated)

AAPL, MSFT, JPM

From

To

2020/01/01

2026/06/03

Return frequency

☒ Daily

☐ Monthly

Fetch data

Default

Das & Statman (2009) base case — 3 securities, pre-calibrated. Reproduces thesis results.

Live market data

Any global ticker from Yahoo Finance — equities, ETFs, indices, crypto (BTC, ETH...)

Enter manually

Enter means, std devs, and correlation matrix directly. Supports 2–10 securities.

Upload CSV

First col = dates, remaining cols = asset prices. Returns computed automatically.

2

Step 2 — Derivative / Structured Product

2
DERIVATIVE / STRUCTURED
PRODUCT

Safety collar (long put + short call) ▼

None — primary securities only

Put option

Call option

Safety collar (long put + short call)

Aggressive collar (long call + short put)

Straddle (long call + long put)

Strangle (long call + long put, diff strikes)

Capital-guaranteed note — uncapped

2
DERIVATIVE / STRUCTURED
PRODUCT

Safety collar (long put + short call) ▼

AI-powered: What is this instrument?

Underlying security
Sec 3 — High risk ▼

Volatility (annualised %)
50.0 - +
Default: 50.0% (std dev of Sec 3 — High risk).
Override for implied vol.

Risk-free rate (%)
3.0 - +

Maturity (years)
0.25 1.00 3.00

Put strike
0.50 1.20 1.50

Call strike
1.00 1.60 2.00

Available instruments

- ▶ Put option / Call option
- ▶ Safety collar (long put + short call)
- ▶ Aggressive collar (long call + short put)
- ▶ Straddle (long call + long put)
- ▶ Strangle (long call + long put, diff strikes)
- ▶ Capital-guaranteed note (capped or uncapped)
- ▶ Barrier-M note
- ▶ Custom composer (calls, puts, digitals, ZCBs)



Select 'None' to optimise without derivatives — Portfolio (1) only

Step 3 — Mental-Account Constraint

3

MENTAL-ACCOUNT CONSTRAINT

Constraint type

☒ VaR — Value at Risk
 ☐ ES — Expected Shortfall

Threshold H (%)

-40
-10
-1

Range extended to -40% to accommodate highly volatile assets (e.g. cryptocurrencies, emerging market equities).

Shortfall probability α (%)

1
5
15

VaR constraint: $P(\text{return} < H) \leq \alpha$

Implied risk-aversion $\lambda = 3.6958$

MV optimal at $\lambda=3.70 \equiv$ behavioural optimal at $H=-10\%, \alpha=5\%$

AI-powered: What is the VaR constraint?

VaR — Value at Risk

$$P(\text{return} < H) \leq \alpha$$

The probability of the portfolio return falling below threshold H must not exceed α . Set H (loss threshold) and α (max probability) using the sliders.

ES — Expected Shortfall

$$E[\text{return} \mid \text{return} < H] \geq L$$

The expected loss in the worst $\alpha\%$ of scenarios below H must not exceed L. Set H, α , and L using the sliders.

💡 Implied λ is computed dynamically — adjust H and α to see the equivalent risk-aversion coefficient update in real time, confirming MVT/MAT equivalence.

4

Step 4 — Grid Resolution

The screenshot shows the 'GRID RESOLUTION' step of the optimiser. At the top, there is a blue header with a white circle containing the number '4' and the text 'GRID RESOLUTION'. Below this is a dropdown menu with a lightning bolt icon and the text 'Fast (m=21, m'=15)'. Under the dropdown is a light blue button with a lightning bolt icon and the text 'AI-powered: What does this resolution mean?'. At the bottom of the interface are two large blue buttons: the top one says '5 ► RUN OPTIMISER' and the bottom one says '↶ Reset / New Simulation'.

Fast (m=21, m'=15)

 ~1–2 min

Quick preview and parameter exploration

Standard (m=35, m'=50)

 ~5–15 min

Good balance — close to thesis results

High precision (m=51, m'=99)

 1–2 hrs

Publication-quality replication of thesis results

Step 5 — Run the Optimiser

✓ **Computation complete** (Total execution time: 1s)

▲ click to collapse

✓ Mean-variance frontier (Execution time: 0s)

✓ Behavioural frontier (no deriv.)

- ✓ ↳ Covariance matrix
- ✓ ↳ State space
- ✓ ↳ Gaussian copula
- ✓ ↳ Grid search
- ✓ ↳ COBYLA refinement — P(1)

✓ Derivative frontier (Execution time: 1s)

- ✓ ↳ Payoff computation
- ✓ ↳ State space (with deriv.)
- ✓ ↳ Gaussian copula
- ✓ ↳ Grid search
- ✓ ↳ COBYLA refinement — P(2)
- ✓ ↳ Portfolio (3) interpolation

✓ Chart rendering (Execution time: 1s)

✓ Results computation (Execution time: 1s)

Computation steps

1. Mean-variance frontier — Markowitz sweep (instant)
2. Behavioural frontier (no derivative) — State space, Gaussian copula, grid search, COBYLA
3. Derivative frontier — Same steps extended with derivative payoff (only if derivative selected)
4. Chart rendering + Results computation

Reading the Chart

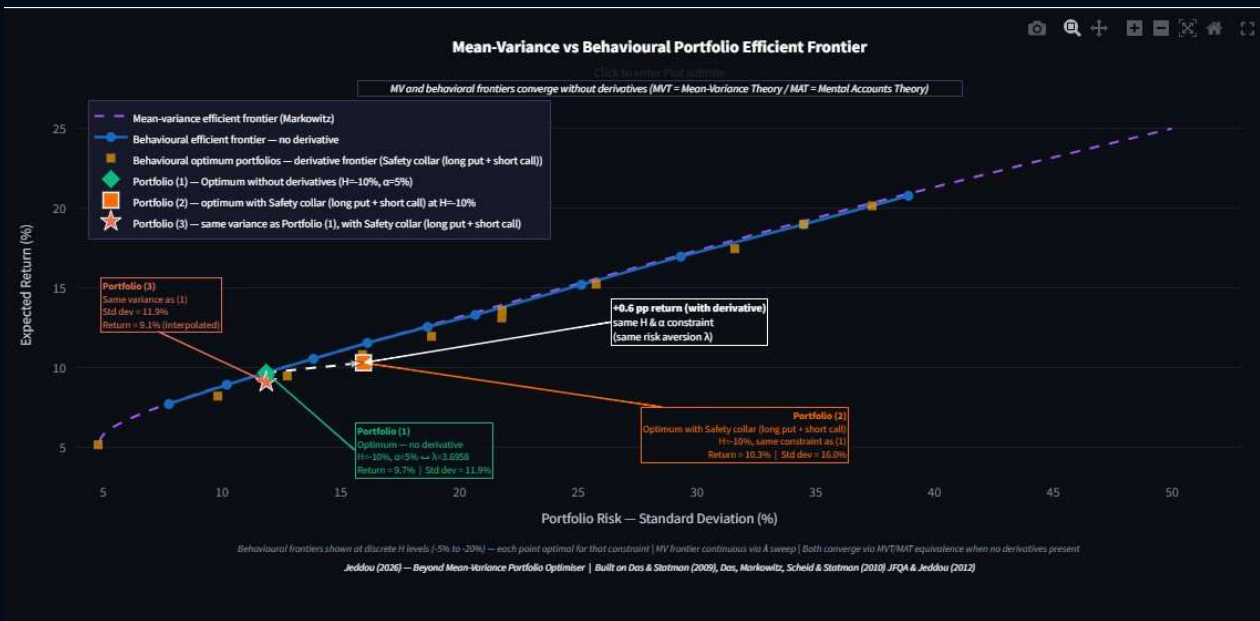
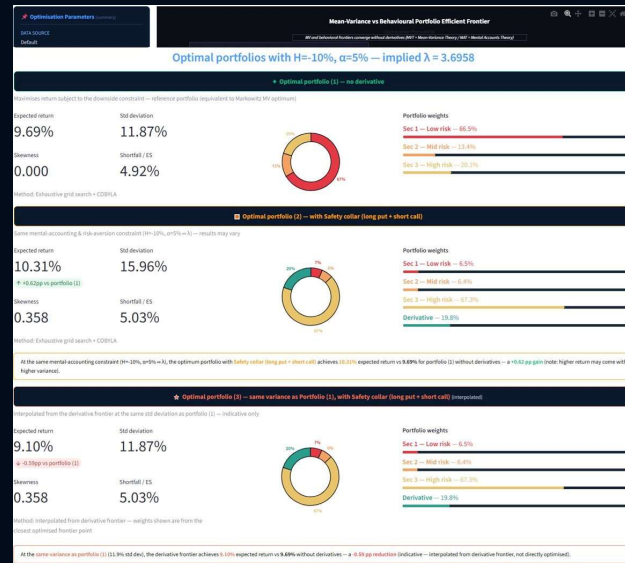


Chart legend

- Purple dashed**
Mean-variance efficient frontier (Markowitz)
- Blue dots**
Behavioural efficient frontier — no derivative
- Gold squares**
Behavioural optimum — derivative frontier
- Green diamond**
Portfolio (1) — no derivative optimum
- Orange square**
Portfolio (2) — with derivative, same λ
- Coral star**
Portfolio (3) — same variance as P(1)

X-axis: Portfolio Risk (Standard Deviation %) | Y-axis: Expected Return (%)

Portfolio Results



Key metrics

Expected return: Annualised mean return of the optimal portfolio

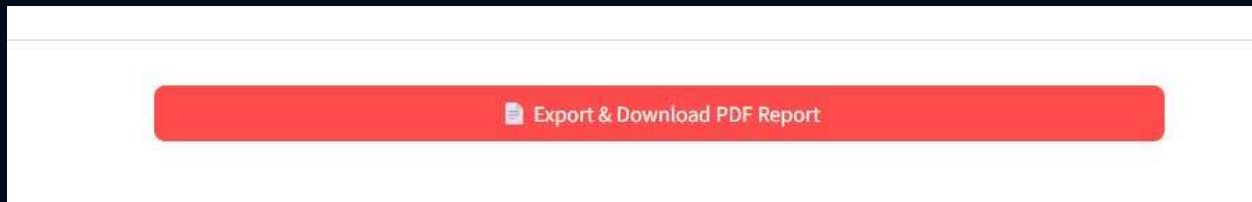
Shortfall / ES: Expected loss in the tail (ES constraint value)

Std deviation: Annualised portfolio volatility (risk measure)

Portfolio weights: Allocation to each security and derivative

Skewness: Asymmetry of return distribution — positive = upside skew

PDF Export



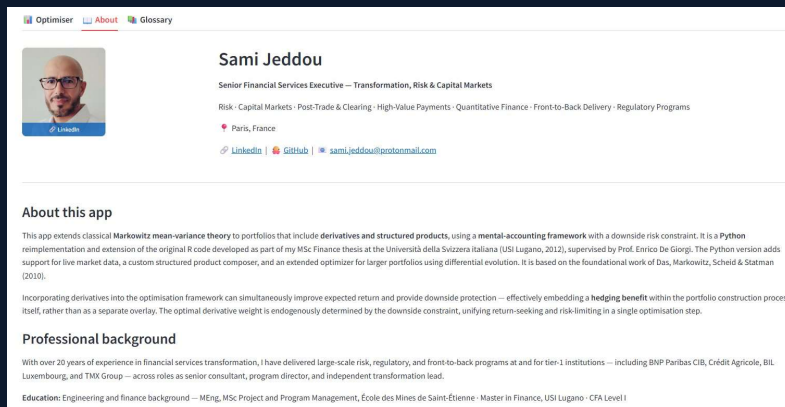
PDF report includes

- ✓ Cover page with run parameters
- ✓ Full efficient frontier chart
- ✓ Portfolio (1), (2), (3) metrics
- ✓ Weight allocation bar charts
- ✓ Legal disclaimer

💡 The PDF is generated in-browser — no data is sent externally. Click 'Download PDF report' after the simulation completes.

About & Glossary Tabs

About Tab



Optimiser About Glossary

Sami Jeddou
Senior Financial Services Executive — Transformation, Risk & Capital Markets
Risk · Capital Markets · Post-Trade & Clearing · High-Value Payments · Quantitative Finance · Front-to-Back Delivery · Regulatory Programs
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About this app

This app extends classical Markowitz mean-variance theory to portfolios that include derivatives and structured products, using a mental-accounting framework with a downside risk constraint. It is a Python implementation and extension of the original R code developed as part of my MSc Finance thesis at the Università della Svizzera italiana (USI Lugano, 2012), supervised by Prof. Enrico De Giorgi. The Python version adds support for live market data, a custom structured product composer, and an extended optimizer for larger portfolios using differential evolution. It is based on the foundational work of Das, Markowitz, Scheid & Statman (2010).

Incorporating derivatives into the optimisation framework can simultaneously improve expected return and provide downside protection — effectively embedding a hedging benefit within the portfolio construction process itself, rather than as a separate overlay. The optimal derivative weight is endogenously determined by the downside constraint, unifying return-seeking and risk-limiting in a single optimisation step.

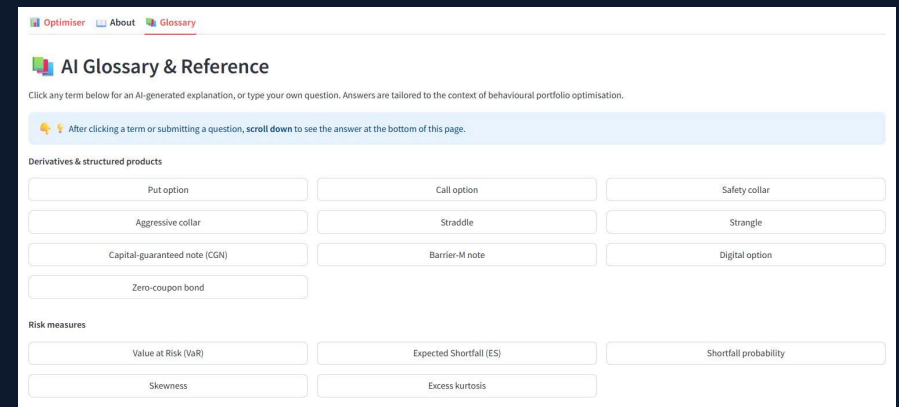
Professional background

With over 20 years of experience in financial services transformation, I have delivered large-scale risk, regulatory, and front-to-back programs at and for tier-1 institutions — including BNP Paribas CIB, Crédit Agricole, BIL Luxembourg, and TMX Group — across roles as senior consultant, program director, and independent transformation lead.

Education: Engineering and finance background — MEng, MSc Project and Program Management, École des Mines de Saint-Étienne · Master in Finance, USI Lugano · CFA Level I

- ▶ Author background & contact
- ▶ App description & algorithm
- ▶ Academic references
- ▶ Legal disclaimer

Glossary Tab



Optimiser About Glossary

AI Glossary & Reference

Click any term below for an AI-generated explanation, or type your own question. Answers are tailored to the context of behavioural portfolio optimisation.

After clicking a term or submitting a question, scroll down to see the answer at the bottom of this page.

Derivatives & structured products

Put option	Call option	Safety collar
Aggressive collar	Straddle	Strangle
Capital-guaranteed note (CGN)	Barrier-M note	Digital option
Zero-coupon bond		

Risk measures

Value at Risk (VaR)	Expected Shortfall (ES)	Shortfall probability
Skewness	Excess kurtosis	

- ▶ Click any term for AI explanation
- ▶ Covers derivatives, risk, portfolio theory
- ▶ Ask your own questions
- ▶ Tailored to behavioural portfolio context

Disclaimer & Contact

Legal Disclaimer

This application is based on the mental accounts portfolio optimisation framework of Das & Statman (2009) and Das, Markowitz, Scheid & Statman (2010), as extended in Jeddou (2012). It is provided for educational and research purposes only and does not constitute financial advice, investment recommendations, or a solicitation to buy or sell any financial instrument. Results are purely illustrative and should not be used as the basis for any investment decision.

Academic References

- ▶ Das, S. & Statman, M. (2009) — Beyond Mean-Variance: Portfolios with Derivatives and Non-Normal Returns in Mental Accounts
- ▶ Das, S., Markowitz, H., Scheid, J. & Statman, M. (2010) — Portfolio Optimization with Mental Accounts, JFQA Vol.45 No.2
- ▶ Jeddou, S. (2012) — Beyond Mean-Variance: Options and Structured Products in Behavioral Portfolios, MSc Finance, USI Lugano